

Use-Case Flow Specification

Problem Statement #34 – Digital Gift Card & Loyalty Point Exchange

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Use Case ID	UC-04
Use Case Name	Redeem Digital Gift Card
Primary Actor	Account Holder
Secondary Actors	Merchant Partner (via UC-03), Fraud Detection System (via UC-06)
Trigger	Account Holder chooses to redeem accumulated exchange credits for a merchant digital gift card.

Preconditions

- Account Holder is authenticated and logged into the digital wallet platform.
- Account Holder holds a non-zero balance of universal exchange credits, previously converted from loyalty points (UC-02).

Postconditions

- A single-use digital gift-card voucher is issued and linked to the Account Holder's wallet.
- The exchange-credit ledger balance is decremented by the redeemed amount with zero balance drift.

Main Success Scenario

1. Account Holder selects “Redeem Gift Card” from the wallet dashboard.
2. System displays the available merchant gift-card catalog with denominations and the exchange-credit amount required for each.
3. Account Holder selects a merchant gift card and a denomination.
4. System validates that the Account Holder's exchange-credit ledger balance covers the requested denomination «include» UC-03 Validate Merchant Loyalty Points.
5. System generates a single-use voucher code with a cryptographic checksum «include» UC-05 Generate Voucher Code.
6. System deducts the redeemed credit amount from the Account Holder's ledger balance with zero balance drift.
7. System displays the gift card voucher code and a confirmation message, and emails a receipt to the Account Holder.
8. Use case ends successfully.

Alternate Flow A – Insufficient Exchange-Credit Balance (extends step 4)

- 4a1. System displays an “Insufficient Balance” message.
- 4a2. Account Holder is redirected to the point-conversion screen (UC-02) to convert additional loyalty points.
- 4a3. If conversion succeeds, the use case resumes at step 3; otherwise the use case terminates.

Alternate Flow B – Suspicious Redemption Detected (extends step 6, «extend» UC-06)

- 6a1. Fraud Detection System flags the redemption as high-risk (e.g., rapid repeated redemptions or a mismatched device fingerprint) «extend» UC-06 Lock Voucher (Anti-Fraud Check).
- 6a2. System places a temporary lock on the generated voucher instead of releasing it.
- 6a3. Account Holder is notified that the voucher is under review, with an estimated resolution time.
- 6a4. If manual review clears the transaction, the voucher is unlocked and step 7 resumes; otherwise the voucher is invalidated and the exchange credits are refunded to the Account Holder's ledger.